MILITARY SPECIFICATION

TRANSMITTING SET, INFRARED AN/SAT-2(

I. SCOPE

1.1 This specification covers Transmitting Set, Infrared, AN/SAT-2( ), an infrared beacon system for installation aboard ship.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-wall

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-S-901 - Shock Tests, H.I. (HIGH-IMPACT). Shipboard Machinery, Equipment and Systems, Requirements for
MIL-D-3464 - Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-Q-9858 - Quality Program Requirements
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore, General Specification (Navy)
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108 - Definition of and Basic Requirement for Enclosure for Electric and Electro ic Equipment
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production
MIL-STD-810 - Environmental Test Methods (with N3020 E2 {11})

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

PUBLIC..:\TIONS

MILITARY

NAVAIR-15-03-500 - Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft

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NAVSHIPS 0967-004-6000 - Instruction Manual for Infrared Transmitting Set AN/SAT-2A

DRAWINGS

9000-S6202-74290 - Rotary Switch, 5-Section, Bulkhead Mounted, Type JRB-B
9000-S6202-F-74140 - Sheet Metal Enclosure - SBM, Size 6- by -9 inch for Standard Electrical Fittings
9000-S6202-F-74272 - Control Key - SBM, for Blinker Lights, Type A-28A, 125-Volt, Base Mounted

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies shall be addressed to the Official Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois, 60606.)

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616P Street, N. W., Washington, D, C, 20036).

3. REQUIREMENTS

3.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4,3 (see 6.3).

3.2 General Requirements. The equipment shall meet the requirements of MIL-E-16400 to the extent specified herein.

3.2 .1 Environmental conditions. The equipment shall be capable of continuous reliable operation under temperature and humidity conditions for class **2** equipment as specified in MIL-E-16400. When subjected to the nonoperating conditions for temperature and humidity for class **2** of MIL-E;.,16400, the equipment shall not be damaged or its performance degraded.

3.2.2 l\1echanical design. The equipment shall conform to the mechanical design requirements of .NIIL-E-16400 regarding construction, welding, soldering, mounting bolts, mechanical operation, shock, vibration, inclination and noise, and accessibility. The equipment shall be submersion-proof in accordanet with MIL-STD-108 to a minimum depth of 3 feet of water. Enclosures shall include stuffing tubes for cablr entrances.

3,2.3 Electrical design. The electrical design of the equipment shall conform to the requirements 01 ::.v.IIL-E-16400 regarding primary power circuits, overload protection, electrical bonding, internal cabling, connecting wire termination, and identification of conductors.

3.2.3.1 Input power. The equipment shall be designed to operate from a 115 volts ±10 percent direc, current (de) or a'll5 volt ±10 percent 60 hertz {Hz) power source.

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3.2.3.2 Circuit isolation. Electric circuits shall be electrically isolated from external parts of the equipment and from internal parts that do not perform circuit functions. Independent circuits shall be electrically isolated.

3.2.4 Parts. Useof parts in the construction of the equipment shall be in accordance with the requirements of MIL-E-16400 regarding selection of parts, design application, surplus parts, cable, capacitors, clamps, terminai boards, terminal strips and connectors, gaskets, switches, and threaded devices.

3.2.4.1 Parts interchangeability. Parts used in the construction of the equipment shall be electrically and mechanically interchangeable with the corresponding parts of the Government furnished model. In the event of conflict between requirements of 3.2.4 and parts interchangeability requirements, parts interchangeability requirements shall be the governing factor.

3.2.5 Materials. The materials used in the construction of the equipment shall be in accordance with the materials section of MIL-E-16400 regarding fungus-inert materials, unacceptable materials, arc-resistant materials, tape lacing, metals, plastics, impregnating, filling, potting and encapsulating compounds **and** electrical tubing and sleeving.

3.2.5.1 Metals. Metal parts shall be made of non-ferromagnetic corrosion resistant materials. Ferromagnetic parts may be used only with written authorization of the command or agency concerned.

3.2.5.2 Gaskets. Gaskets, with the exception of those used in the infrared transmitter, shall be in accordance with the requirements of MIL-E-16400. Gaskets used in the infrared transmitter shall be made of silicone rubber of a type approved by the command or agency concerned. Other features of the infrared (IR) transmitter gasket design shall conform to the requirments of MIL-E-16400.

3.2.6 Coating and treatments. Coating and treatments shall be in accordance with the coating and treatments provisions of MIL-E-16400.

3.2.7 Safety design. Safety design shall be in accordance with requirement 1 of MIL-STD-454.

3,2.8 Thermal design. Thermal design considerations shall conform to applicable sections of MIL-E-1o-100.

3,2.9 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454,

3.2.10 Controls, indicators, and panel layout. Use of controls, indicators, and panel layout shall be consistent with requirements of MIL-E-16400.

3,2,11 Supplementary requirements. Supplementary requirements of MIL-E-16400 regarding serial numbers, date of manufacture, instruction plates, Government nomenclature, and identification and markings shall be complied with unless otherwise specified in the contract or purchase order.

3.2.12 Visual security. There shall be no white light leaks from any part of the equipment, with the exception of the transmitting beacon when it is operating.

3.3 Detailed requirements.

3.3.1 Functional description. The equipment shall be functionally and operationally equivalent to thE AY-/SAT-2A infrared transmitting set described in NAVSHIPS 0967-004-6000 and applicable drawings listP• herein. It will permit one set of manual keys to be used in either the circuit of the infrared transmitting set AX/SAT-2A, or of its visible counterpart - the Yardarm Blinker, but will not permit inadvertent opera tion of these visible lights when they are secured for darkened ship conditions in the usual way. The bea<'n'_ shall be operable, separately or together, by either of the manual keys and the selector switch,

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3.3.2 Equipment composition. The AN/SAT-2A infrared transmitting set shall consist of the following:

(a) Infrared transmitter (beacon)	**2** each
(b) Transmitter control	**1** each
(c) Telegraph key	**2** each
(d) Spare lamps	4 each
(e) Technical manuals	**2** each

3.3.3 Weights and dimensions. The maximum dimensions and weights of the equipment units shall be as follows:

Name of unit	Overall dimensions - inches			Weight - pounds
	Height	Weight	Depth	
Infrared transmitter (beacon)	13-1/4	12-1/2	12-1/2	20-3/4
Transmitter control	10-3/4	8	6-3/8	10-1/4
Telegraph key	5-7/8	6-3/4	6-3/4	6

3.3.4 Infrared transmitter. The infrared transmitter (beacon) shall be as described in the Government furnished microfilm drawings and NAVSHIPS 0967-004-6000 except as otherwise specified herein. The major components of the m transmitter shall include the infrared filter and the transmitter lamp.

3.3.4.1 Infrared filter. Theinfrared filter shall consist of a homogeneous, heat treated glass dome.

3.3.4,l.l Visual and infrared transmission. The IR filter shall limit the visual transmission of the IR transmitter while permitting transmission of infrared energy. For a 2850° Kelvin tungsten source the glass filter shall have a visual transmission (EVT) of not greater than 4X10-6 as measured on a high density photometer and an infrared transmission (EHT) of not less than 25 percent as measured on an infrared filter photometer utilizing a receiver with an S-1 spectral response (see 4,5,4). Alternatively, measurements may be obtained on a visible-near infrared spectrophotometer and converted to EVT - EHT values.

3,3,4.1.2 Solar radiation. Visual and infrared transmission characteristics of the filter shall not be degraded by exposure solar radiation of the intensity normally encountered in an outdoor shipboard environment.

3.3.4.l.3 Mechanical and thermal strength. The infrared filters shall be properly constructed and processed to assure adequate mechanical and thermal strength. Each filter shall be capable of withstanding the breakage test of 4.5.10, the thermal strength test of 4.5.6, the thermal shock test of 4.5.7, without damage and the impact test of 4.5.3.3.

3.3.4.1.4 Filter uniformity. The filter characteristics test shall be conducted to assure dimensional and optical uniformity and to eliminate defects from all filters and domes delivered. Tests shall include surface inspection, weight, dimensions and visual transmission. The visual transmission test shall be conducted in a darkened area. Thefilter and dome shall be illuminated from within by a light bright enough to reveal any flaws, seeds, bubbles, cracks, or thin spots in the glass. The filter and dome shall be rotated and inspected from all angles, Any void, bubble, or seed exceeding 1/8 of an inch in diameter, or a total of defects sufficient to cover 10 percent of the cylindrical surface of the filter, shall be unacceptable.

3.3.4.2 IR transmitter lamp. The transmitter lamp shall be a 115 volt, 300 watt, 200 hours rated-life :nedium bipost base, T-20, clear bulb with two heavy-duty coiled tungsten filaments connected in parallel. It shall be equivalent to and interchangeable with I 101 of NAVSHIPS 0967-004-6000.

3.3.4.3 Reflector. The reflector shall have a spherical inner surface of bright specular aluminum. Reflectivity of the inner surface shall be not less than 85 percent.

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3,3.4.4 Intensity distribution. The peak light intensity of the IR transmitter with a new lamp and with the IR filter removed shall be 1250 candelas ±20 percent. The beam spread to fifty percent of the peak intensity shall be 180° ±10° in a horizontal plane passing through the lamp filament. The beam spread to fifty percent of peak intensity in a vertical plane perpendicular to the beam axis shall be 40° above the horizontal and 23° below the horizontal ±10 percent. The intensity distribution of the lamp and reflector combination shall be approximately as shown in figure 1.

3,3.5 Transmitter control assembly. The transmitter control assembly shall consist of a selector switch, panel type fuse holders, indicator lights and necessary terminal strips. Controls shall be waterproof and shall be mounted on the cover of a standard 6 by 9 inch sheet metal enclosure conforming to 9000- S6202-F-74140.

3,3,5.1 Selector switch, The selector switch shall be a type 3-JRB-5S switch conforming to 9000- S6202-74290 modified for panel mounting and conforming with 3.2,5,1. An identification plate indicating switch positions shall be attached to the panel concentric with the switch shaft as illustrated in NAVSHIPS 0967-004-6000. An external locking device shall not be required.

3,3,5,2 Fuse holders. Fuse holders shall be blown fuse indicating, panel mounting waterproof cartridge types and of a type approved by the command or agency concerned, Both lines to each beacon shall be fused with 6 ampere fuses.

3.3,5,3 Current indicators. Each current indicator shall light when current is being drawn by the lamp to which it pertains. It shall be a dual lamp fixture filled with a red Polaroid attenuator. Voltage for the indicator lamps shall be derived from the voltage drop across a 0.7 ohms ±10 percent, 20-watt wire wound resistor in series with the beacon lamp. The indicator shall be a 2.5 volts, 0.50 ampere, tungsten filament lamp **with** a rated life of more than 25 hours. The attenuator shall be continuously variable between complete darkness and maximum intensity.

3.3.5.4 Identification plates and control nomencla.ture. Since this equipment may be installed with the beacons either forward and aft or on the port and starboard sides.of the ship, interchangeable markings shall be furnished for all controls so that the beacons **will** be correctly identified in the event of either type of installation.

3,3,5,5 Wiring diagram and operating instructions. A copy of the wiring diagram of the equipment shall be secured within each control unit. Brief installation notes and operating instructions shall be included. The printed matter shall occupy both faces of a white card laminated in vinyl or similar preservative material.

3,3,6 Manual key. The manual telegraph key shall be as shown on 9000-S6202-F-74272 but modified to include a red indicator light. It shall be submersion proof to a depth of 3 feet of water in accordance with ::.\UL-STD-108 and provided with a hinged protective cover. The key shall be provided with a quick-hold-down quick-release mechanism to provide control, at the key, for blinker or steady operation. Material shall conform to 3,2.5.1.

3.3.6.1 Manual key capacitor. The key shall be provided with a 2.0 microfarad, 150 working volts capacitor to be connected across the key contacts. Bince the capacitor is used only when the system is operated from a de source, the capacitor shall be furnished mounted in place but with the leads disconnected and the lugs taped.

3.3 ,6.2 Indicator. The indicator shall light when the key is closed or when the switch is in the "Steady Operation" position. The light shall be an indicator lamp equivalent to part I 201 of NAVSHIPS 096 7-0G4-6000,

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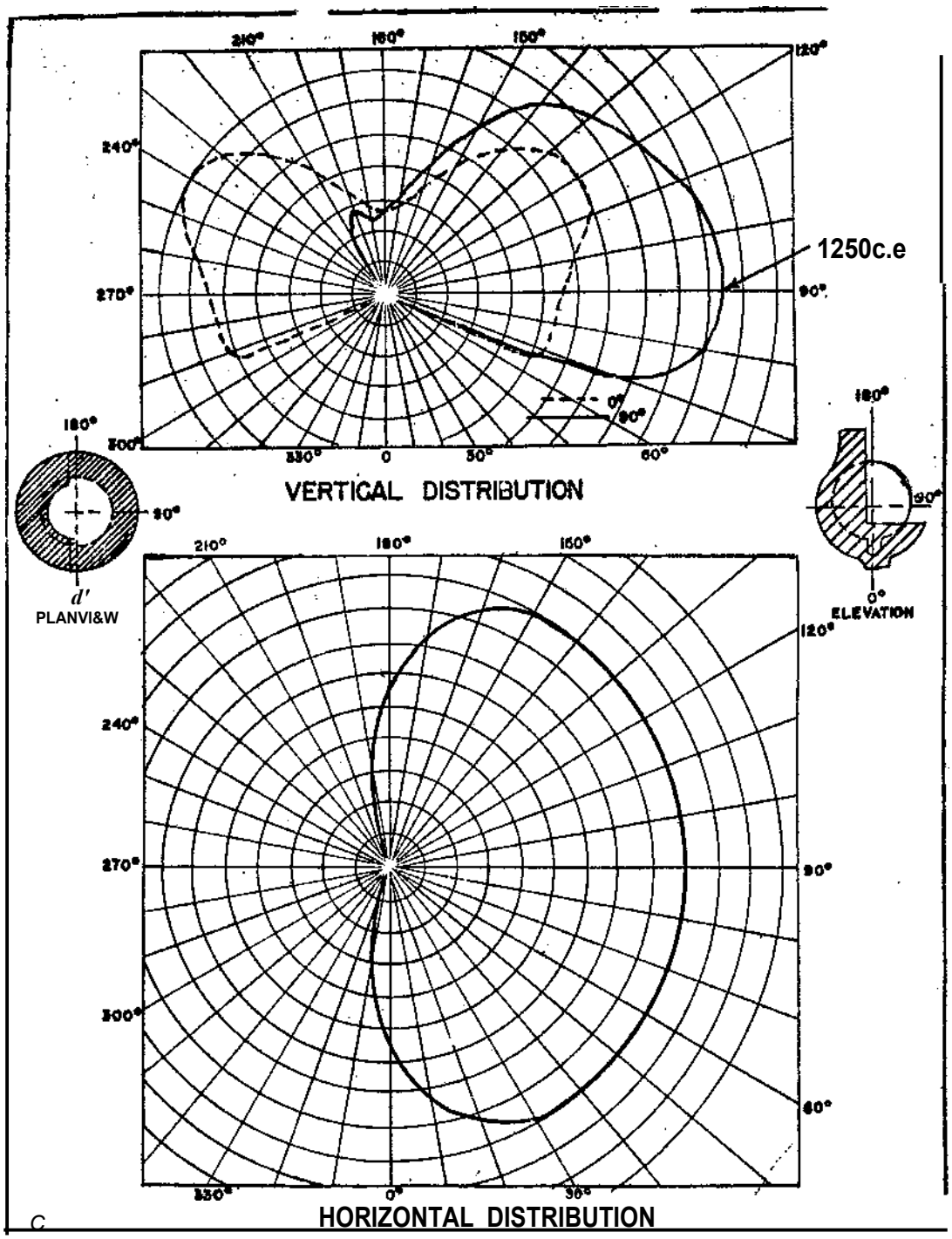


FIGURE 1. Radiation intensity distribution for AN/SAT-2( ) with IR filter removed.

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3.4 Reliability

3.4.1 Mean-time-between-failures. The mean-time-between-failures for this equipment, excluding lamps, shall be 5000 hours.

3.4.2 Reliability program plan. The supplier shall establish and implement a reliability program in accordance **with** MIL-STD-785 (see 6.2).

3.4.3 Reliability prediction. The supplier shall perform a reliability prediction in accordance with the methods of MIL-HDBK-217 entitled "How to Make a Reliability Prediction Based on Stress Factors and Part Population• (see 6.2).

3.4.4 Failure rates. Theprimary source of failure rate data shall be MIL-HDBK-2 l 7. For those components not listed in MIL-HDBK-217, other sour,::es of failure rate data may be used with the prior approval of the command or agency concerned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.l Quality program requirement. The supplier shall provide and maintain a quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858.

4.1.1.l Quality plan. A quality plan, applicable to the requirements of this specification shall be submitted to the Quality Asirnrance Representative for approval prior to first article inspection (see 6.2). If first article inspection is not required the quality plan shall be submitted for approval prior to quality conformance inspection. The quality plan shall reflect the following:

(a} flow of material from inspection on receipt through all manufacturing processes to final shipment
{b) process and inspection points
(c)) reference to process and inspection procedures
(d) differentiation between in plant inspections and subcontracted inspections for subassemblies and higher assemblies.

Any changes to the approved plan shall be submitted to the Quality Assurance Representative for approval prior to use.

4.1.2 Government verification. All quality assurance operations performed by the supplier will *be* subjected to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b} Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the supplier to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures. The supplier shall submit written test procedures, including test equipment and tolerance limits, to be used for first article inspection and quality conformance inspection to the command or agency concerned via the Government inspector for review and approval prior to beginning tests (see 6.2). Each test shall be identified in accordance with 4.2 as to classification.

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4.1.3.1 Procedures for inspections, tests, demonstrations, and so forth shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name, model, range and accuracy. Special test equipment shall also be listed and identified.

4,1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

(1) Production inspection (see 4.4.1)
(2) Production control inspection (see **4.4.2)**
(3) Environmental inspection (see 4.4.3)
(4) Reliability testing (see 4.4.8)

(c) Inspection of preparation for delivery (see 4.6)

4.3 First article inspection. Unless otherwise specified (see 6.1), two units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

TABLE I. First article inspection

Examination or Test	Requirement Paragraph	Test Method Paragraph
Surface examination	3.2.4, 3.2.5, 3.2.6, 3.2.9, 3.2.11, 3.3.4, 3.3.4.1.4, 3.3.5, 3.3.6	4.5.2
Operating test	3.2.3.2, 3.2.7, 3.2.12, 3.3.1, 3.3.4.1.3, 3.3.5.3, 3.3.6.2	4.5,3
Weights and dimensions	3.3.3	4.5.1
Supply line voltage and frequency	3.2-.3.1	4.5.1
Heat test	3.2.8	4.5.1
Enclosure test	3.2.2	4.5.1
Temperature test	3.2.1	4.5.1
Humidity test	3.2.1	4.5.1
Visual and IR transmission	3.3.4.1.1	4.5.4
Intensity distribution	3.3.4.4	4.5,5
Controls and control circuits	3.2.10, 3.3.6.1, 3.3.6.2	4.5.1
Thermal strength (IR filter)	3.3.4.1.3	4.5.6
Thermal shock (IR filter)	3.3.4.1.3	4.5.7
Shock vibration inclination, noise and structural integrity	3.2.2	4,5.1
Salt spray test	3.2.5.1	4.5.1
Magnetic test	3,2.5.1	4.5.1
Weld test	3.2.2	4.:i.1
Solar radiation (IR filter)	3.3.4.1.2	4.5.8

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Table I. First article inspection (Continued)

Examination or Test	Requirement Paragraph	Test Method Paragraph
Reflectivity (reflector)	3,3.4.3	4,5.9
Breakage test (IR filter)	3,3.4.1.3	4,5.10
Interchangeability	3,2.4.1	4.5,11

4.3.1 First article inspection report. A first article inspection report containing all the results of the first article inspection shall be prepared (see 6,2). The completed Form NAVEXOS 3373 as specified in MIL-S-901 for shock test and resilient mount data sheet as specified in MIL-E-16400 shall be included in the report along with evidence of the ability of the equipment to perform its intended function during and after shock and vibration tests. First article approval will not be granted unless the report is complete. Any production entered into prior to approval of the first article will be at the suppliers risk,

4.4 Quality conformance inspection.

4.4,1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such e,;amination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4,1.3 and shall include the following:

TABLE II. Production inspection

Examination or Test	Requirement Pa_ragraph	Test Method Paragraph
Surface examination	3,2.4, 3.2.5, 3.2.6, 3.2.9, 3.2.11, 3.3.4, 3.3.4.1.4, 3.3.5, 3,3.6	4.5.2
Operating test	3,2.3.2, 3,2.7, 3.2.12, 3,3.1, 3,3,4.1,3, 3,3.4.2, 3.3,5,3, 3.3.6,2	4,5,3

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

TABLE III. Production control inspection

Examination or Test	Requirement Paragraph	Test Method Paragraph	AQL (percent defective)
Weights and dimensions	3.3.3	4.5.1	6,5
Supply line voltage and frequency	3,2,3.1	4.5.1	6.5
Heat test	3.2.8	4,5.1	6.5
Enclosure test	3.2.2	4,5.1	6,5
Visual and IR transmission	3,3.4.1.1	see 4,5,4	
Intensity distribution	3.3,4,5	4,5,5	6.5
Controls and control circuits	3.2.10	4.5.1	6.5

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Table III. Production control inspection (Continued)

Examination or Test	Requirement Paragraph	Test Method Paragraph	AQL (percent defective)
Thermal strength (IR filter)	3.3.4.1.3	4.5.6	6,5
Thermal shock (IR filter)	3.3.4.1.3	4,5.7	6.5
Magnetic test	3.2.5.1	4.5. 1	6,5
Reflectivity (reflector)	3.3.4.3	4.5.9	6.5
Breakage test	3.3.4.1.3	4.5.10	

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection. Environmental inspection shall be conducted to prove the durability of the equipment to withstand the effects of extremes of environment, service conditions, and hard usage. Environmental inspection shall be performed on equipment(s) that have passed production inspection as specified in **4,4.1** and shall include the tests listed in table IV.

TABLE IV. Environmental inspection

Examination or Test	Requirement Paragraph	Test Method Paragraph
Temperature test	3.2.1	4,5.1
Humidity test	3.2.1	4.5.1
Solar radiation	3.3.4.1.2	4.5.8
Shock, vibration		
Inclination, noise	3.?.2	4.5.1

4.4.3.1 Environmental inspection samples. Samples for the environmental inspection shall be selected without regard to their quality in accordance wit!:. the following:

(a) One sample from the first 35production units fabricated.
(b) One sample from each successive 100 units produced, or fraction thereof.

4.4.4 ::\'on-conformance and retest. If a sample unit fails a test specified in 4.4.2, 4.4.3, and 4.4.8, the supplier shall immediately investigate the cause of failure and shall report to the Government inspec or the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes; and so forth. The equipment shall then be retested,

4.4.5 Corrective action. If corrective action results in a change of part(s); electrical and mechanical layout or function; or if the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the supplier cannot determine the cause of failure, the matter shall be referred immediately to the contracting officer. It shall then be determined by the command or agency concerned whether correction should be made to the equipments already delivered under the contract,

4.4.6 Reinspection of conforming production control and environmental test units. Unless otherwise :::;pecified, sample units which have been subjected to and passed production control and environmental inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

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4.4.7 Quality conformance inspection report. The supplier shall submit to the Government a report detailing the results of all testing performed (see 6.2). The report shall list the serial numbers of equipments tested during the reporting period. During the conduct of tests, performance parameters shall be measured and verified in accordance with this specification. The report shall include completed Form NAVEXOS 3373 for the shock test and, when resilient mounts are used, compliance with figure 3 of lVIIL-E-16400. Equipment shall not be accepted unless the report is complete.

4.4.8 Reliability testing.

4.4.8.1 Qualification testing. The first eight (8) production units shall be selected for testing in accordance with MIL-STD-781, test level D, test plan VI, except that voltage cycling is not required. Qualification testing is not required when there is no requirement for first article inspection.

4.4.8.2 Production testing. Every 3rd month eight (8) equipments shall be randomly selected by the government inspector for testing in accordance with 4.4.8.1 herein.

4.4.8.3 ReliabHity test procedures. Reliability test procedures for qualification and production testing shall be in accordance with the "Detail Requirements" section of MIL-STD-781. Reports on reliability testingare required in accordance with MIL-STD-781 (see 6.2).

4.5 Inspection procedures.

4.5.1 Tests and examinations. All tests and examinations specified herein shall be conducted in accordance **with** the applicable inspection procedures of MIL-E-16400, except as follows:

4.5.2 Surface examination. In addition to the surface examination procedu:r-es of MIL-E-16400 to demonstrate conformance to 3.2.4, 3.2.5, 3,2.6, 3.2..9, 3.Z.11, 3,3,4, 3.3.5, and 3.3,6, the equipment shall be subjected to an optical uniformity inspection of the filter and dome. The optical inspection shall consist of surface and through transmission visual examinations for compliance to 3.3.4.1.4. The through transmission inspections shall be conducted in a darkened area using sufficiently intense back illumination to reveal any internal defects. The parts shall be rotated and examined from all angles.

4.5.3 Operating test. Theequipment shall be energized and subjected to an operating test to insure the proper functioning and performance at an adequate number of points in the required range to demonstrate conformance to specification requirements, including safety. The operating test shall include, but not be limited to, the tests shown in table V.

TABLE V. Operating test

Test	Requirement Paragraph
Functional/ performance	3,2.7, 3.3.1, 3,3,5,3, 3.3.6.2
Circuit isolation	4,5.3.1
Visual security	4.5,3.2
Impact test (IRfilter)	4.5.3.3
Beam intensity (lens)	4.5.3.4

4.5.3.1 Circuit isc-lation. A potential of 1200 volts, 60 Hz, shall be applied continuously for one minute between circuits and between each circuit and the metal external surface of the equipment to demonstrate conformance to 3.2.3,Z, Current leakage shall not exceed 100 microamperes. This potential shall not be applied across lamps and capacitors.

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4.5.3.2 Visual security. The equipment shall be set up in a completely darkened area and the power turned on. The control panel. switches, and beacons shall be examined for white light leaks to demonstrate conformance to 3.3.4.1.4. There shall be no white light emitted from any part of the equipment with the exception of void or bubbles allowable under 3.3.4.1.4. A uniform red glow transmitted through the filter and dome is a normal condition and is not cause for rejection.

4.5.3.3 Impact test. Each IR filter shall withstand the following test or an equivalent test approved by the command or agency without cracking, chipping or breaking to demonstrate compliance with 3.3.4.1.3. The filter shall be placed open end down on the stage of a suitable pendulum type impact machine against and within the opening of a 90° Vee back-stop, the face of which is 1/4 inch thick pressed wood, or equivalent, of Rockwell hardness 60 to 95 on the "R" scale. Before testing, the backstop and freely hanging hammer shall be centered 2 inches above the stage. The line of action shall biset the backstop. The hammer shall consist of a 2 inch plastic (phenolformaldehyde formula 175) ball, having a Rockwell hardness of 105 to 120 on the "M" scale, which shall be mounted in a steel cage. The filter and dome shall be subjected to a two foot-pound pendulum type impact at three points located 120° apart around the circumference without breage.

4.5,3.4 Beam intensity. Photometric measurements shall be made to determine compliance with **.3.4.Z.**

4.5.4 Visual and infrared transmission test. Three two inch by two inch square samples shall be taken from each melt of glass used in the manufacture of the infrared filters, The samples shall have approximate thicknesses of 25 percent, 50 percent 100 percent of the nominal thickness of the IR filter. The samples shall be used for determination of compliance with visual and IR transmission requirements of 3.3.4.1.l. The visual and infrared transmissions of each sample shall be measured as described in 3.3.4.I.l and forwarded to the command or agency concerned for approval.

4.5.5 Intensity distribution. The IR transmitter with a new lamp and with the IR filter removed shall be tested for intensity distribution for conformance to 3,3.4.4.

4.5.6 Thermal strength test. TheIRfilter shall be held for a period of Z4 hours at an ambient temperature of +Z00°F and allowed to cool normally in free air to demonstrate compliance with 3.3.4.1.3. There shall be no fractures during any part of this test.

4.5.7 Thermal shock test. The IR filter shall be heated uniformly to a temperature of 295°F, then suddenly and completely, or partly, immersed in water at 45°F without breaking to demonstrate conformance to 3.3.4,l.3.

4.5,8 Solar radiation test. TheIRfilter shall be subjected to the sunshine test of MIL-STD-810, test method 505, procedure I for conformance to 3,3,4.1.2. Any change in the IR or visual transmission characteristics of the filter as a result of the sunshine test shall be considered as unacceptable.

4.5.9 Reflectivity test. The inner surface of the reflector shall be tested for conformance to 3.3.4.3.

4.5.10 Breakage test. A suitable sample of parts selected from each lot of IR filters processed, through the strain relieving operation, shall be subjected to the following breakage test. TheIRfilter shall be broken with a standard steel center punch applied at the apex of the dome or two inches up the cylindrical side, *The* punch shall be struck hammer blows of increasing severity until the energy of the final blow is just sufficient to cause breakage. The IR filter when broken shall disintegrate into pieces, none of which shall be larger than an inch on the longest dimension. The sampling procedure and the lot accept and reject criteria for this process control test shall be included in the test procedures submitted in accordance with 4.1.3.

4.5.I1 Interchangeability. Interchangeability shall be demonstrated to show compliance to 3.2.4.1.

4.6 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5.

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5, PREPARATION FOR DELIVERY

{The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by suppliers shall be specified in the individual order.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified **(see 6.I).**

5,1,1 Level A.

5.1.1.1 Cleaning. Cleaning shall be in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Drying shall be in accordance with MIL-P-116.

5,1.1.3 Preservative application. None.

5.1.1,4 Unit packaging. Each infrared transmitting set shall be individually packaged in accordance with MIL-P-116, insuring compliance with the General paragraph under Methods of Preservation {Unit Protection) and the Physical Protection paragraph therein. Cushioning and dunnage shall be used to prevent physical or mechanical damage.

5.1.1.4.1 Major components. The infrared transmitters {beacon) shall be packaged in accordance with Method IIb of MIL-P-116. Desiccant shall conform to MIL-D-3464, Type II or III.

5.1.1.4.2 Other components. The transmitter control, manual telegraph keys, and spare lamps shall be packaged in accordance with Method III of MIL-P-116 using containers conforming to PPP-B-636.

5,1.1.4.3 Optical parts. Optical parts shall be preserved and packaged in accordance with MIL-O-16898.

5,1.1.4.4 Equipment repair parts. Equipment repair parts or running spares shall be individually packaged in sealed transparent packages in accordance with MIL-E-17555. These items shall be packaged separately from other items within the pack in containers conforming to PPP-B-636 and identified to the applicable equipment. Repair parts lists shall be provided in accordance with MIL-E-17555,

5,1.1.4.5 Technical manuals. Technical manuals shall be packaged in a waterproof, transparent, heat sealed bag, minimum 4 mil thick, and shall not be placed within the barrier used to package any other items.

5,1.2 Level C. Each complete infrared transmitting set shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified {see 6.1).

5.2.1 Level A. Items comprising a complete infrared transmitting set, shall be packed in fiberboard containers conforming to PPP-B-636, V Grade, Special Use Requirement, for Type 2 Load. In lieu of the closure and waterproofing requirements in the appendix to PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, corners, and manufacturer's joint with waterproof tape, 2 inches minimum width, conforming to PPP-T-60, Class 1, or PPP-T-76. Reinforcing shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2 .2 Level B. Items packaged as specified in 5.1, shall be packed in containers conforming to PPP-B-636, class weather-resistant in a satisfactory condition at the lowest applicable rate. Containers, packing, or method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules, or regulations of other carriers, as applicable to the mode of transportation.

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5.3 Unitized loads. When applicable, items packed as specified in 5.2 shall be palletized or containerized in accordance with MIL-STD-147 whenever quantities to one destination equal 40 cubic feet or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated in shipping containers specified in 5.2. Palletized loads shall be strapped to the pallet. Unitized loads shall not be subjected to rough handling tests. When applicable, items for shipboard end destinations shall be unitized by hull number.

5.4 Marking. In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with MIL-STD-129 (see 6.1)." Marking shall include special marking for Method II packs.

5.5 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible. Unless otherwise specified, loose-fill polystyrene shall not be used for packaging or packing, Adhesives shall not be used in the closure of wraps, bags, or boxes.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Levels of preservation-packaging and packing, and applicable marking (see section 5).
(c) Number of first article samples to be submitted if other than specified in 4.3.

6.2 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423). The data required by this specification include, but are not restricted to the following:

(a) Reliability program plan (see 3.4.2)
(b) Reliability prediction (see 3.4.3)
(c) Quality plan (see 4.1.1.1)
(d) Test procedures (see 4.1.3)
(e) First article inspection report (see 4.3.1)
(f) Quality conformance inspection report (see 4.4.7)
(g) Reliability test procedures (see 4.4.8.3)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy-EC

(Project 5850-::\030)

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